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TCP Connection

(Server
receives
the request
in form of
packets).

① Packets reaches the NIC on the server side.

② NIC has internal queue called ring buffer to store temporarily those packets.

① NIC raises the interrupt.

② Interrupt is handled by
ISR (Interrupt Service
Routine) on first half.

③ ISR schedules softirq.
(second half).

① Softirq handles processing of packets in form of batches. (It can process millions of packets in batches)

② Softirq send the processed packets to TCP/IP stack.

② TCP/IP :-

1. validate checksum.

2. Ordering.

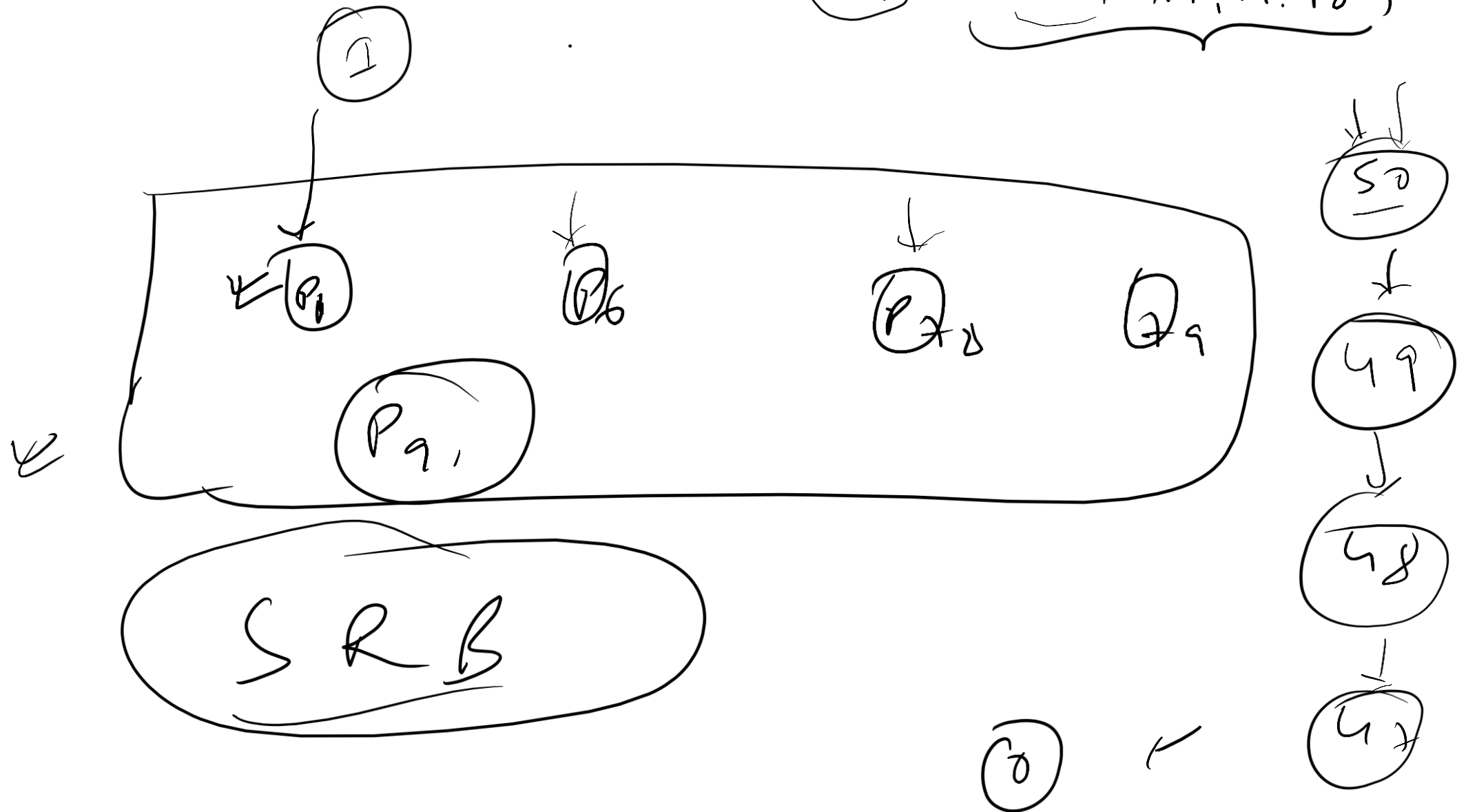
3. Reassembly

③ TCP/IP layer will send it
to socket receive Buffer

① epoll will get to know that there is a packet which is ready to be read from socket receive buffer. (SRB)

② read command in redis which will read all the files or packets present in SRB.

$$R_1 \rightarrow \underbrace{(P_1, P_{1.1}, P_{1.10})}$$



① Redis runs a single-threaded event loop as follows :-
while (true)

ready_clients = poll(epoll-fd)

for client in ready_clients :

read_command(client)

queue_command_for_execution(client)

execute_next_command_in_queue()